

江苏大学全英文授课精品课程

Arc Length

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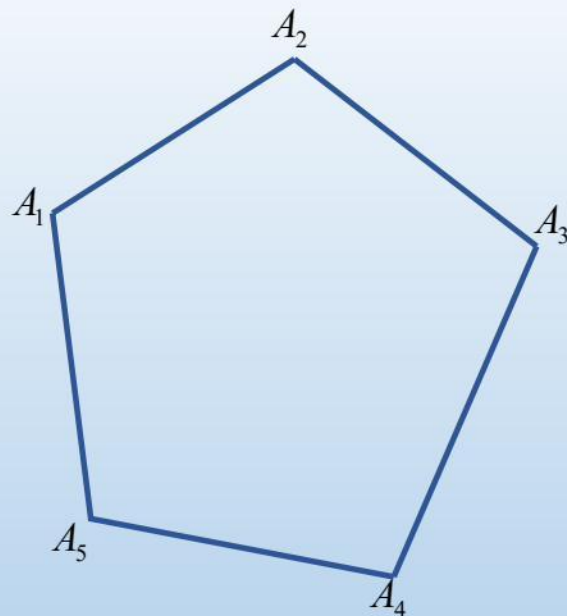
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1 What's the Meaning of The Length of a Curve?



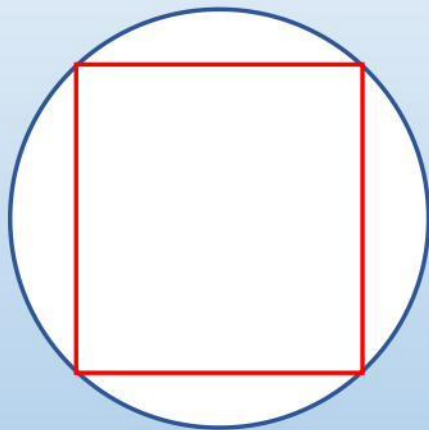


length of the polygon: $|A_1A_2| + |A_2A_3| + |A_3A_4| + |A_4A_5| + |A_5A_1|$

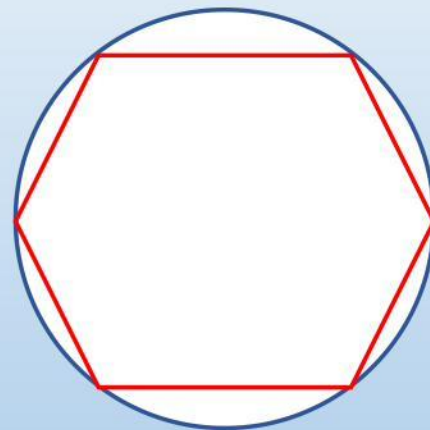




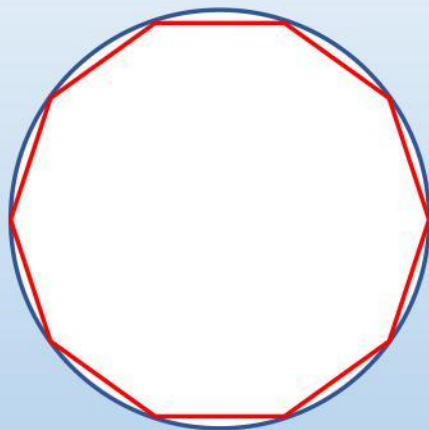
The case of a circle:



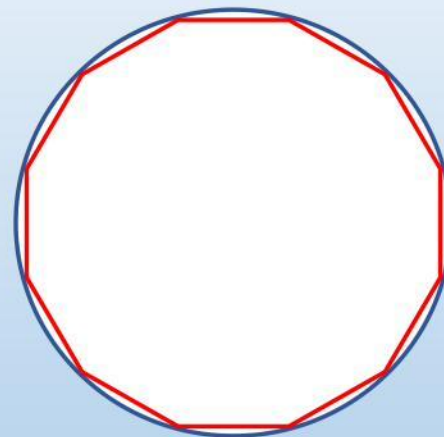
The number of segments: $n = 4$



The number of segments: $n = 6$

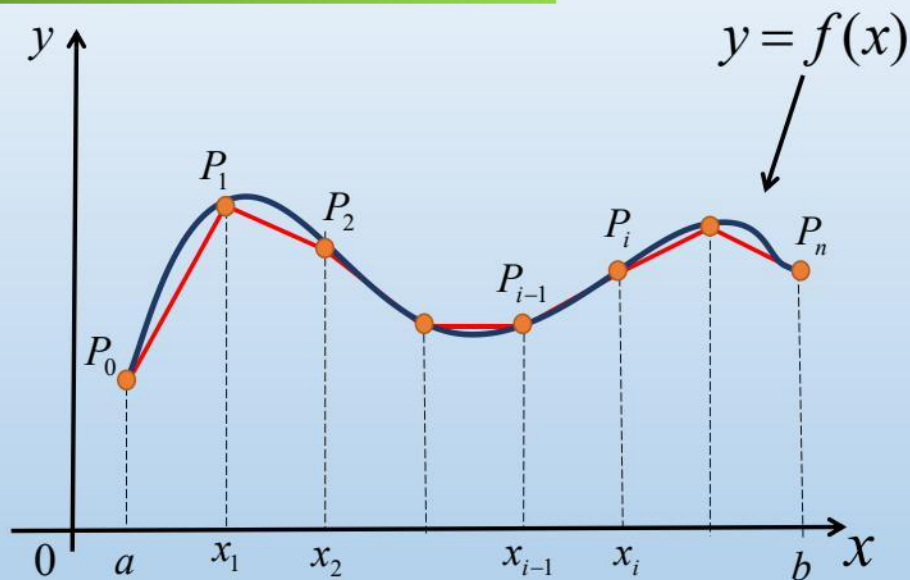


The number of segments: $n = 10$



The number of segments: $n = 12$

The length of a general curve:



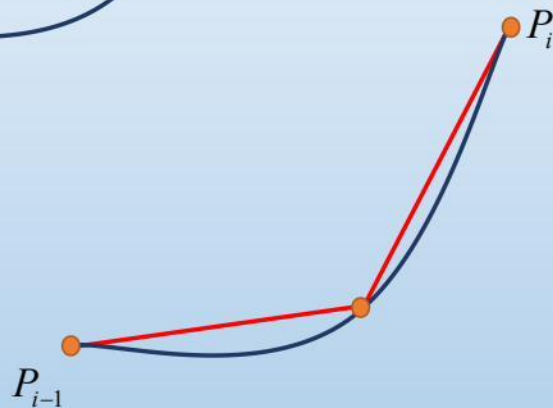
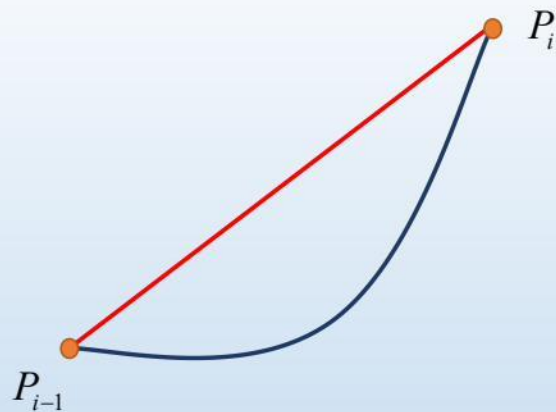
$$L \approx |P_0P_1| + |P_1P_2| + \dots + |P_{n-1}P_n| = \sum_{i=1}^n |P_{i-1}P_i|$$



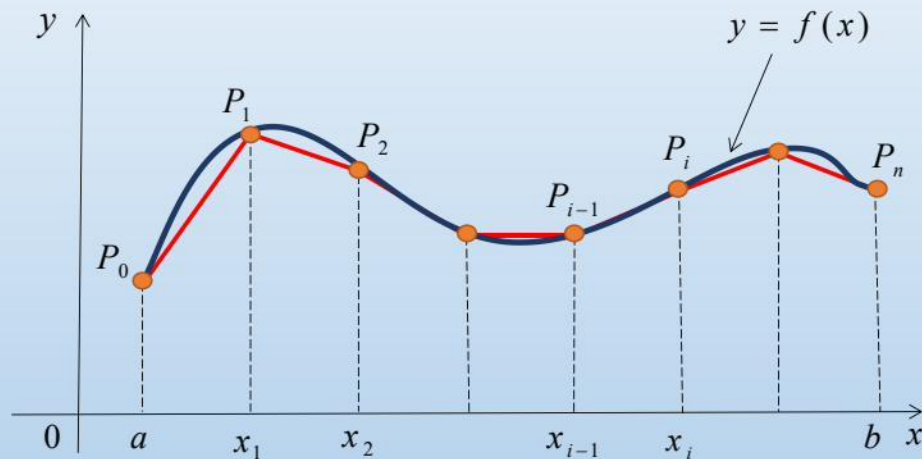


More Segments

More Accuracy



Definition of the length of curve $y=f(x)$:



$$L \approx |P_0P_1| + |P_1P_2| + \dots + |P_{n-1}P_n|$$
$$= \sum_{i=1}^n |P_{i-1}P_i|$$

Definition :

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i|$$



2 The Arc length Formula

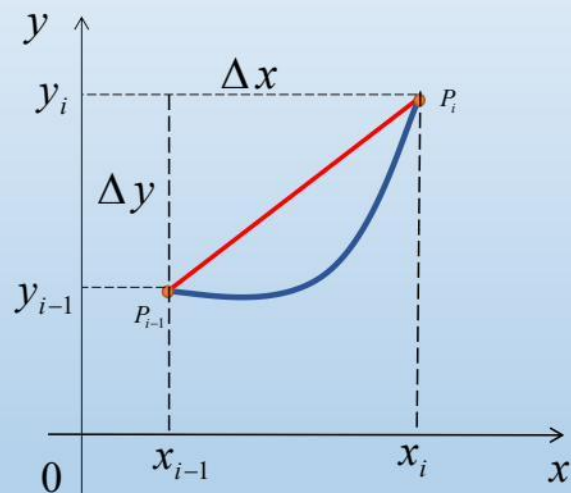
Definition :
$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i|$$

- * First, we divided the curve into a large number of small parts.
- * Then, we found the approximate lengths of small parts and add them.
- * Finally, we took the limit as $n \rightarrow \infty$.

Definite Integral:
$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i)\Delta x_i$$



Definition: $L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i|$



$f(x) \in C'[a, b]$

$$|P_{i-1}P_i| = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

$$\Delta y_i = f(x_i) - f(x_{i-1}) = f'(x_i^*) \cdot \Delta x_i$$

$$|P_{i-1}P_i| = \sqrt{(\Delta x_i)^2 + [f'(x_i^*) \cdot \Delta x_i]^2} = \sqrt{1 + [f'(x_i^*)]^2} \cdot \Delta x_i$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i| = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + [f'(x_i^*)]^2} \cdot \Delta x_i \\ &= \int_a^b \sqrt{1 + [f'(x)]^2} dx \end{aligned}$$

This integral exists!



The Arc Length Formula:

If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$, $a \leq x \leq b$, is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Remark:

- * Determine the function of the curve, the independent variable and the interval.
- * Evaluate the derivative of the function.
- * Use the arc length formula to calculate.

If $x = g(y)$ and $y \in [c, d]$, then the length of the curve $x = g(y)$, $c \leq y \leq d$, is

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy$$





Example 1 Find the length of the arc of the semicubical parabola $y^2 = x^3$ between the points (1,1) and (4,8).

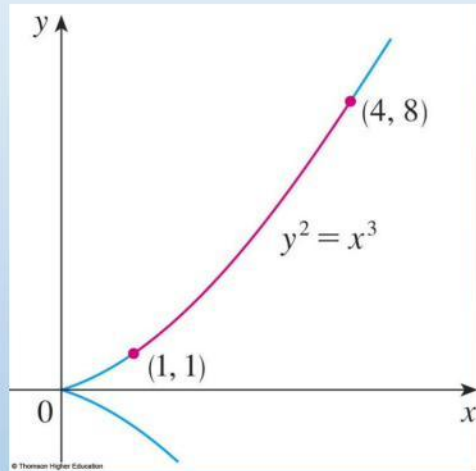
Solution: For the top half of the curve, we have

$$y = x^{\frac{3}{2}} \quad \text{and} \quad y' = \frac{3}{2} x^{\frac{1}{2}}$$

independent variable is x , and $x \in [1, 4]$

So according to the formula, the arc length is

$$\begin{aligned} L &= \int_1^4 \sqrt{1 + (y')^2} dx = \int_1^4 \sqrt{1 + \frac{9}{4}x} dx \\ &= \frac{1}{27} (80\sqrt{10} - 13\sqrt{13}) \end{aligned}$$



u-substitution $u = 1 + \frac{9}{4}x$

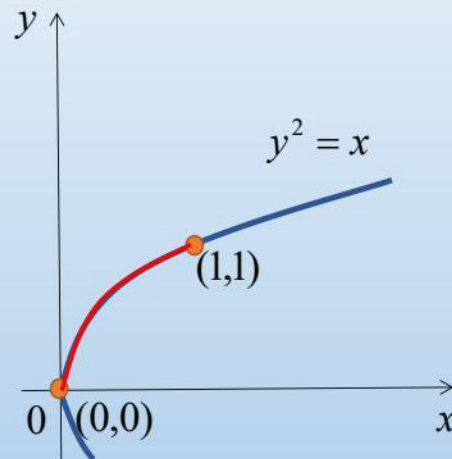
Example 2 Find the length of the arc of the parabola $y^2 = x$ from $(0, 0)$ to $(1, 1)$.

Solution: Since $x = y^2$, we have $\frac{dx}{dy} = 2y$

Here independent variable is y , and $y \in [0, 1]$

So according to the formula, the arc length is

$$\begin{aligned} L &= \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_0^1 \sqrt{1 + 4y^2} dy \\ &= \frac{\sqrt{5}}{2} + \frac{\ln(\sqrt{5} + 2)}{4} \end{aligned}$$



trigonometric substitution $y = \frac{1}{2} \tan \theta$

Example 3 Find the circumference of a circle with radius R .

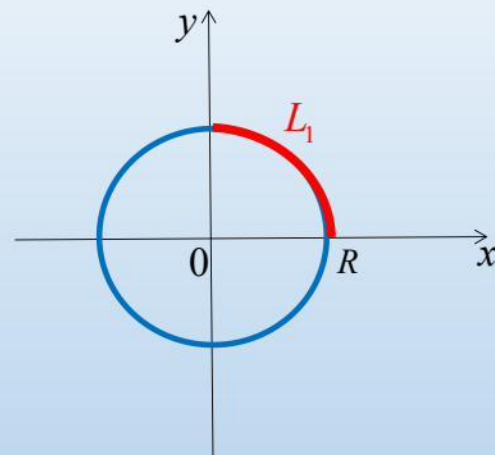
Solution: Place the circle at the coordinate system,
the origin is at the center of the circle,
and the x-axis passes through the diameter.

The function of this circle is $x^2 + y^2 = R^2$

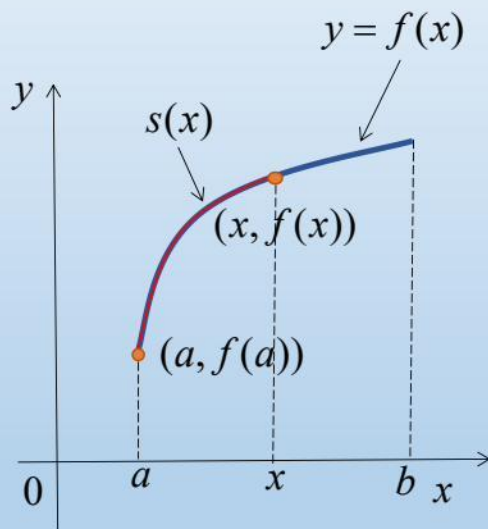
$$L_1: y = \sqrt{R^2 - x^2}, \quad 0 \leq x \leq R \quad y' = \frac{-x}{\sqrt{R^2 - x^2}}$$

$$\text{So, } L_1 = \int_0^R \sqrt{1 + y'^2} dx = \int_0^R \sqrt{1 + \frac{x^2}{R^2 - x^2}} dx = \frac{1}{2} \pi R$$

$$L = 4L_1 = 2\pi R$$



3 The Arc length Function



$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt = \int_a^x \sqrt{1 + [f'(t)]^2} dt$$

$\Downarrow$ FTC1

$$\frac{ds(x)}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}$$

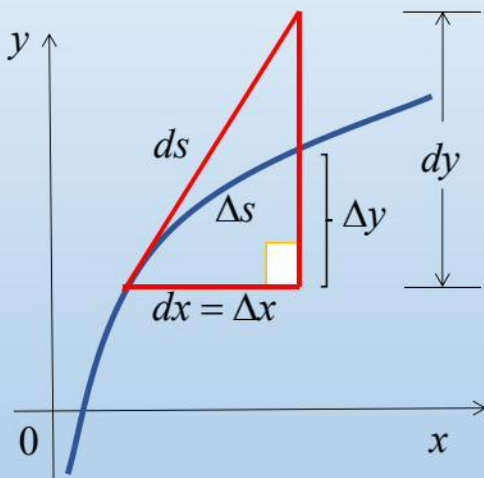
$\Downarrow$

$$ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx$$

$\Downarrow$

$$(ds)^2 = (dx)^2 + (dy)^2$$

Geometric Interpretation:



$$ds = \lim_{\Delta x \rightarrow 0} \Delta s = \lim_{\Delta x \rightarrow 0} \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$(ds)^2 = (dx)^2 + (dy)^2$$





Example 4 Find the arc length function for the curve $y = x^2 - \frac{1}{8} \ln x$ taking

$P_0 (1, 1)$ as the starting point.

Solution: Since $f(x) = x^2 - \frac{1}{8} \ln x$, we have $f'(x) = 2x - \frac{1}{8x}$

$$1 + [f'(x)]^2 = 4x^2 + \frac{1}{64x^2} + \frac{1}{2} = \left(2x + \frac{1}{8x}\right)^2$$

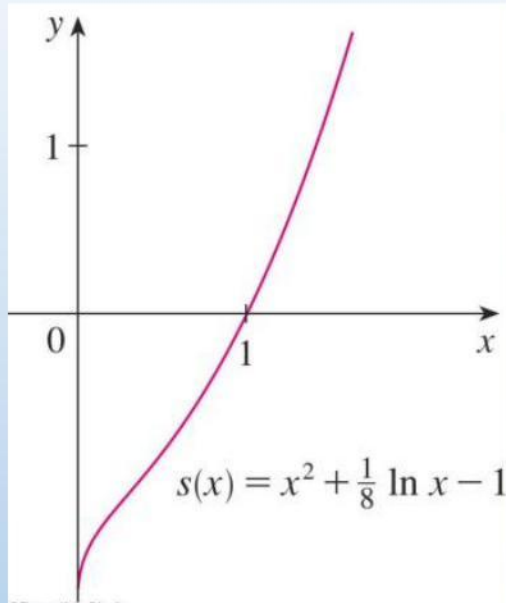
So according to the formula, the arc length function is

$$s(x) = \int_1^x \sqrt{\left(2t + \frac{1}{8t}\right)^2} dt = \int_1^x \left(2t + \frac{1}{8t}\right) dt$$

For instance, the arc length along the curve from (1,1) to (3,3) is :

$$s(3) = 3^2 + \frac{\ln 3}{8} - 1 = 8 + \frac{\ln 3}{8} \approx 8.1373$$





Why is $s(x)$ negative when x is less than 1?





Exercises 1

Find the exact length of the curve.

(a) $y = 1 + 6x^{\frac{3}{2}}, \quad 0 \leq x \leq 1$

(b) $x = \frac{1}{3}\sqrt{y}(y-3), \quad 1 \leq y \leq 9$

(c) $y = \ln(\cos x), \quad 0 \leq x \leq \frac{\pi}{3}$

(d) $y = \frac{1}{2}x^2, \quad \text{From } P - \frac{1}{2} \text{ to } Q(1, \frac{1}{2})$

